



**Verified Carbon
Standard**

INSTALLATION OF HIGH EFFICIENCY WOOD BURNING COOKSTOVES IN RWANDA

Document Prepared by

C-Quest Capital Stoves Asia Limited

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Prepared By	C-Quest Capital Stoves Asia Limited
Contact	Address: C-Quest Capital Stoves Asia Limited, Brumby Centre, Lot 42, Jalan Muhibbah, 87000 Labuan F. T., Malaysia. Telephone: +1-202 247-797 Email: cqc-operations@cquestcapital.com Website: www.cquestcapital.com

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1 PROJECT DETAILS

1.1 Summary Description of the Project

A summary description of the technologies/ measures to be implemented by the project.

The project involves distribution of fuel-efficient improved cookstoves (ICS) in Rwanda. The ICS (TLC-CQC Rocket Stoves) disseminated through this project will replace the baseline cookstoves.

Through this project, the distribution and installation of approximately 500,000 ICS will be undertaken for households in Rwanda. Under this project, Total Land Care Rocket Stoves, henceforth referred as TLC-CQC Rocket Stoves, will be installed per household in Rwanda. Each household will receive maximum two units of TLC-CQC Rocket Stoves.

The ICS will burn wood more efficiently, hence saving fuel. Not only will this halt the rapidly progressing deforestation in Rwanda but will also reduce health hazards from indoor smoke pollution and women and children will be able to spend less time collecting firewood.

In this grouped project, each ICS constitutes a single project activity instance which is in line with the definition stated in VCS Program Definitions version 04.2. The end user will be informed in advance that the use of ICS generates carbon finance which in turn is used for recovering the price of ICS and project implementation costs.

As on 30-April-2022, 9,334 ICS have been deployed in Rwanda.

The location of the project.

All the project activity instances in the project activity would be in geographical boundaries of Rwanda. The details of the project location are provided in Section 1.12.

An explanation of how the project is expected to generate GHG emission reductions or removals.

The ICS will substitute the currently common cooking on open fire. The ICS burns wood more efficiently, hence saving fuel and lowering greenhouse gas emissions.

A brief description of the scenario existing prior to the implementation of the project

The baseline scenario is the continued use of non-renewable wood fuel by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity.

An estimate of annual average and total GHG emission reductions and removals.

The annual average GHG emission reduction from the project is expected to be 3,393,437 t CO₂e.

The total GHG emission reductions achieved for 10 years fixed crediting period is expected to be 33,934,377 t CO₂e.

1.2 Sectoral Scope and Project Type

The project is categorised under type/category as below:

a) Sectoral scope: 03 - Energy demand

b) Type: II – Energy efficiency improvement projects

This project is a grouped project.

1.3 Project Eligibility

The project involves energy efficient cookstove distribution which falls under the category of efficiency improvements in thermal applications, therefore it is eligible under the scope of VCS Program.

1.4 Project Design

The project is a grouped project in this grouped project, each ICS constitutes a single project activity instance which is in line with the definition stated in VCS Program Definitions version 4.2

Eligibility Criteria

For the inclusion of new project activity instances, the project proponent shall ensure that it meets the eligibility criteria below.

No.	Criterion	Compliance Requirement	Method of Evaluation
1	Methodology: Meet the applicability conditions set out in the methodology applied to the project	The project activity instance shall use VCS approved methodology-VMR0006: Methodology for Installation of High Efficiency Firewood Cookstoves, Version 1.1 and shall meet the applicability conditions set out in the methodology.	Details of how the new project activity instances (ICS) meet the applicability conditions can be confirmed in Section 3.2 of this VCS-PD.
2	Technology: Use the technologies and measures specified in the project description	The project activity instance shall implement TLC-CQC Rocket Stoves having minimum efficiency of 25%; below which it shall not be eligible under this grouped project.	Manufacturer 's specification shall be used for establishing minimum efficiency criteria in the first year of stove installation.

3.	Baseline Scenario: Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area.	All new project activity instances will be installed within Rwanda only and the baseline scenario is the continued use of non-renewable wood fuel by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity. This also conforms to the applied methodology.	Baseline scenario can be confirmed in Section 3.4 of this VCS-PD. Baseline scenario and geographical boundary can be confirmed from registration records cum consent deeds of individual ICS which contains geographical coordinates, address as well as confirmation by end user (stove owner) stating that they were using traditional three stone fire or pot support before the installation of project stove.
4.	Additionality: Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area.	Each project activity instance shall demonstrate its compliance with- 1. Regulatory Surplus: The project should not be mandated by any law, statute or other regulatory framework, or for UNFCCC non-Annex I countries, any systematically enforced law, statute or other regulatory framework 2. Positive List: The inclusion of new project activity instances shall have to comply with positive list that is the ICS are installed/ distributed at zero cost to end user and the project activity instance has no other source of revenue but GHG credits.	There is no law, statute or Government programme or policy in Rwanda mandating the project activity nor is there any systematically enforced law, statute, or other regulatory framework for such projects. Hence this requirement is satisfied for all project activity instances to be included in the current grouped project Registration form cum consent deed or any other documentary evidence mentioning that end user were not charged for the ICS. Same can be confirmed during verification.
5	Occur within one of the designated geographic areas specified in the project description	All new project activity instances will be installed within Rwanda only.	Project Activity Instance Database containing Geographical Coordinates of each household to which cookstoves are distributed.

6	Ownership: Have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance	A default Beneficiary Agreement for end users including the provision that emission reductions generated by the project activity are owned by the Project Proponents will be provided for project activity instance.	Copy of registration form cum Consent deed of end user(Stove owner) to Project Proponents regarding emission reduction claims.
7	Start Date	The start date of project activity instance shall be same as or later than the grouped project start date.	Project Activity Instance Database containing date of installation of each ICS.
8	Where a capacity limit applies to a project activity included in the project, no project activity instance shall exceed such limit.	1. The aggregate energy savings by a single project activity shall not exceed the equivalent of 180 GWh_{th}/year as mentioned in the methodology AMS II G: Energy efficiency measures in thermal applications of non-renewable biomass. 2. Each project activity instance that exceeds one percent of the capacity limit (i.e., 1.8 GWh_{th}) shall be identified. Such instances shall be divided into clusters, whereby each cluster is comprised of any system of instances such that each instance is within one kilometer of at least one other instance in the cluster. Instances that are not within one kilometer of any other instance shall not be assigned to clusters.	No project activity instance shall exceed the applicable limit, which is 180 GWh_{th}/y 1. Project Activity Instance Spreadsheet containing estimation of annual energy saving from each project activity instance (i.e. each project ICS). 2. Project Activity Instance ER Spreadsheet containing estimation of annual energy saving from each project activity instance (i.e., each project ICS). The expected annual energy saving for each project activity instance (i.e., each project ICS) comes out to be 0.2288 GWh_{th}/ TLC Rocket Stove well below 1% (i.e., 1.8 GWh_{th}) of capacity limit. Therefore, the project activity instances under this grouped project shall not be assigned to clusters.

9.	Double Counting	The Improved Cookstove distributed under this project shall be uniquely identifiable based on the distribution records.	Project Activity Instance Database containing unique serial number of each ICS. Each ICS distributed will have corresponding end user (Stove owner) details (i.e., name, geographical coordinates, address, unique serial number). The combination of this unique serial number of each ICS along with geographical address of the stove owner confirms the unique identity of each project device
10	Target Population	End user for each project activity instance shall be households, with non-renewable biomass on inefficient firewood stoves. Each ICS will be assigned a unique serial number with name of ICS user, address, GPS of household, stove model, distribution date, etc	Database/ICS Registration form if distribution has started or Self-declaration by project participant.
11	Crediting period	Be eligible for crediting from the start date of the instance through to the end of the project crediting period (only). Note that where a new project activity instance starts in a previous verification period, no credit may be sought for GHG emission reductions or removals generated during a previous verification period (as set out in Section 3.4) and new instances are eligible for crediting from the start of the next verification period.	The start date of the instance will be the date, the stove began generating GHG emission reductions or removals. The end of the crediting period of instance will be the end date of project crediting period as mentioned in section 1.9.

1.5 Project Proponent

Organization name	C-Quest Capital Stoves Asia Limited
Contact person	Ken Newcombe
Title	Director
Address	Brumby Centre, Lot 42, Jalan Muhibbah, 87000 Labuan, Malaysia.
Telephone	+1-202 247-797
Email	cqc-operations@cquestcapital.com

1.6 Other Entities Involved in the Project

Apart from the Project Proponent, there are no other entities involved in the Project.

1.7 Ownership

The right of ownership is with C-Quest Capital Stoves Asia Ltd., this can be confirmed from the consent letter/records of consent deed signed by end user (stove owner) to project proponent during the registration process.

This is in line with clause 5 of section 3.6.1. of VCS project standard version 04.3

1.8 Project Start Date

09-October-2021 (date of commissioning of 1st ICS under this grouped project).

1.9 Project Crediting Period

09-October- 2021 to 08-October-2031, ten years fixed crediting period

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals of the project are:

Project Scale	
Project	
Large project	✓

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
Year 2021-22 (09-October-2021 to 08-October-2022)	1,246,056
Year 2022-23 (09-October-2022 to 08-October-2023)	2,459,313
Year 2023-24 (09-October-2023 to 08-October-2024)	3,576,315
Year 2024-25 (09-October-2024 to 08-October-2025)	4,603,241
Year 2025-26 (09-October-2025 to 08-October-2026)	4,308,977
Year 2026-27 (09-October-2026 to 08-October-2027)	4,033,240
Year 2027-28 (09-October-2027 to 08-October-2028)	3,774,877
Year 2028-29 (09-October-2028 to 08-October-2029)	3,532,806
Year 2029-30 (09-October-2029 to 08-October-2030)	3,306,011
Year 2030-31 (09-October-2030 to 08-October-2031)	3,093,541
Total estimated ERs	33,934,377
Total number of crediting years	10
Average annual ERs	3,393,437

1.11 Description of the Project Activity

The project involves distribution of fuel-efficient improved cookstoves (ICS) to replace the baseline cookstoves in households. It is envisaged that TLC-CQC Rocket Stove (two cookstoves per household) will be installed under this grouped project.

TLC-CQC Rocket Stove usage substantially reduces fuel consumption and emissions for conducting cooking and water heating tasks in homes. The ICSs are more efficient than traditional firewood stove as they reduce heat loss and improve thermal transfer and combustion efficiency. TLC -CQC Rocket Stoves are fuel efficient, resulting in a decrease in fuel use in comparison with traditional firewood stove. It substantially reduces wood fuel consumption compared with a three-stone fire or traditional pot support.

Technology/Measure

TLC-CQC Rocket Stove uses a total of 15 bricks that will be made by the households using locally available clay. The average size of the brick would be 22cm x 11cm x 6.5cm. The bricks will be joined together using a mixture of 3 litres clay, 3 litres sand, 3 litres manure/cow dung and 5 litres of water. This ensures reduction in heat loss and better insulation. Metal components have been added to the design to optimize combustion and heat transfer.



TLC-CQC Rocket Stove

Stove components

The Metal top of the stove improves airflow into the combustion chamber and out the top of the stove by allowing the pot to sit higher. An adjustable metal pot skirt ensures a better transfer of heat from the fire to the pot, improving efficiency and reducing wind resistance.

TLC-CQC stoves come with a metal stick support which is placed in front of and slightly into the opening of the stove and acts as a firewood feeding platform. This ensures adequate airflow while feeding the fuel into the combustion chamber resulting in complete combustion of wood.

According to independent stove efficiency tests performed by Aprovecho Research Centre on the TLC Rocket Stove, the WBT results yielded an average thermal efficiency of 34.5% for boiling of water.

Technical Specifications	
Stove Size	Depth: 34.5 cm
	Width: 34 cm
	Height: 28 cm
Efficiency	34.50%
Average Life	10 years

The life of TLC stove is at least 10 years owing to it being a clay and brick stove which can be easily repaired using a mixture of local clay, cow dung, sand and water on a regular basis. The metal parts used in the stoves are under extended warranty from PP/manufacturer side

throughout the projected life span and will be replaced in case of request received from the stove owner for replacement. PP has confirmed the life span of 10 years through the above measures.

Data collection of ICS end-user

Project proponent must gather the necessary information to identify households using its ICS during the course of the project. To facilitate this process, each ICS will be assigned a unique serial number. This number will be recorded during the registration process together with the following information (as appropriate and as available):

- Name of ICS user or head of the household
- Address / GPS of ICS household
- Phone number of ICS user or household, where available.
- Stove model
- Date of distribution/installation
- ICS serial number
- Distributer information

The information collected will be stored in the electronic database which will serve as project database for project monitoring and sampling purposes.

Ex-ante v/s Actual emission reductions

The ex-ante emission reductions have been calculated as demonstrated in section 4.4. However, the actual emissions reductions during the verification will depend on the values of $B_{y=1,new,i,j,survey}$ and $N_{y,i,j}$ obtained during the monitoring surveys. Also, the projected stove installation might get extended over a year or maybe even two years depending on the rate of construction of the stoves on field and therefore for first few years the emission reduction could be lower than the ex-ante calculated values. The uncertainty in number of stoves installed and operational during the monitoring period will primarily result in the differences of the ex-ante and actual emission reductions achieved during the crediting period.

1.12 Project Location

The project location will be the geographical boundary of Republic of Rwanda with coordinates 1°56'25" S latitude and 29°52.433' E longitude.¹

Below are the map of Rwanda and geographical boundaries.

¹ <https://www.geodatos.net/en/coordinates/rwanda>



Rwanda map²



Map of project area

² <https://en.wikipedia.org/wiki/Rwanda#/media/File:RwandaGeoProvinces.png>

1.13 Conditions Prior to Project Initiation

The conditions prior to project initiation are the continued use of non-renewable wood fuel (firewood) by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity. The pre-project devices are inefficient three-stone fires or conventional systems lacking improved combustion air supply mechanism and flue gas ventilation system i.e., traditional stoves.

Pre project scenario can be confirmed in Section 3.4 of this VCS-PD.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

There are no laws and regulations governing the use of improved cookstoves in Rwanda households. The project is a voluntary effort by the project proponent.

A review is made on Rwanda environmental laws and regulations as below:³

- a) Law No. 04/2005
Organic Law determining the modalities of protection, conservation and promotion of environment in Rwanda
- b) Ministerial Order No. 003/2008
Ministerial Order relating to the requirements and procedure for environmental impact assessment
- c) Ministerial Order No. 004/2008
Ministerial Order establishing the list of works, activities and projects that have to undertake an environment impact assessment
- d) General guidelines and procedure for environment impact assessment

Article 71 from Organic Law:

Any activity aiming at controlling soil erosion and drought, one that aims at afforestation and forestry, using renewable energy in a sustainable manner, using modern cooking stoves and any other means that can be used to protect forestry, may receive support from the National Fund for Environment.

There is no negative concern made on improved cookstoves project from the above laws and regulations.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project has not been registered, nor is it seeking registration under any other GHG program.

1.15.2 Projects Rejected by Other GHG Programs

The project has not been rejected by any other GHG program.

³ <https://elaw.org/rwanda-environmental-impact-assessment-law-and-regulations>

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

The project is not included in an emissions trading program or any other mechanism that includes GHG allowance trading.

1.16.2 Other Forms of Environmental Credit

The project has not sought or received another form of GHG-related environmental credit.

1.17 Sustainable Development Contributions

In accordance with Paragraph 3.17.1 of VCS Standard, version 4.3, “Where project complete a validation or verification to the Sustainable Development Verified Impact Standard (SD VISTA) Program at the same time as a VCS Program validation or verification, they are not required to conduct a separate demonstration of compliance with the requirements set out in this Section 3.17”

Status of the project under SD Vista : PP has submitted the documents to Verra for the listing. Therefore, details of SDGs claimed is not required to be mentioned in VCS-PD⁴

1.18 Additional Information Relevant to the Project

Leakage Management

Not applicable as the project adopts a net gross adjustment factor of 95% to account for leakage.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

The project contributes to sustainable development in a number of ways:

a) Environmental Sustainability

- The project will help significantly reduce greenhouse gas emissions over its lifetime.

⁴ In accordance with the paragraph 3.17.1 of VCS Standard, version 4.3

- The project will help reduce the use of non-renewable biomass from forests, thus assist in conserving existing forest stock and the protection of natural forest eco-systems and wildlife habitats.

b) Social Sustainability

- Considerably less time need to be spent collecting wood fuel for the family home thereby reducing the work burden on rural families and presenting alternative opportunities for economic development.
- The amount of indoor pollutants from the burning of biomass in the family home will be reduced. Less carbon dioxide, carbon monoxide and particulates will be emitted due to the decrease in total biomass burned and an increase in the temperature of combustion.
- The stove provides a safer method for combusting biomass for cooking, helping to reduce burn injuries, especially for children, in the family home.

c) Economic Sustainability

- The project will help develop a section of the local economy, in the distribution, local assembly, maintenance and monitoring activities.
- Household expenditures on cooking fuel will be reduced through the use of the ICS.
- Saved household labor can be diverted to more productive economic activities.
- The project will create local employment opportunities in operational and management roles, as well as future assembly and/or manufacturing initiatives.

2 SAFEGUARDS

2.1 No Net Harm

No potential negative environmental or socio-economic impacts have been identified for the project. The project activity promotes environmental and socio-economic wellbeing. Also, project activity generated local employment which supports upliftment of socio-economic status of region

2.2 Local Stakeholder Consultation

Feedback was requested from local stakeholders for the “Installation of high efficiency wood burning cookstoves in Rwanda” program between 26-October-2020, and 18-December-2020. Necessary precautions were taken place in light of the COVID-19 pandemic, with feedback having been conducted electronically when possible, minimizing large groups and socially distancing during in person meetings.

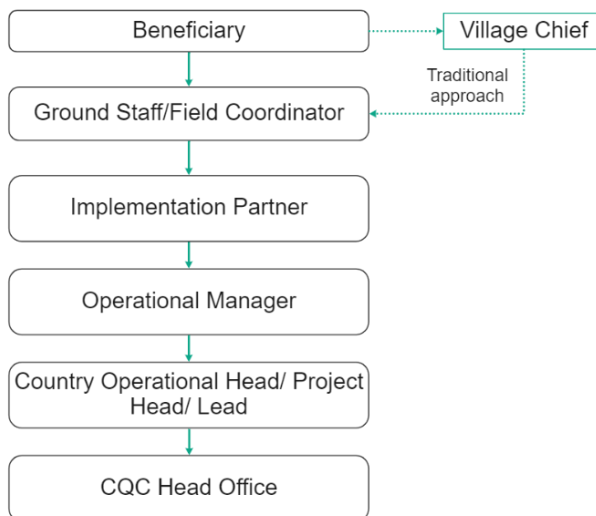
Stakeholders were invited to provide feedback via multiple methods including an announcement posted on C-Quest Capital’s website on 26-October-2020 with an electronic feedback form, Project Description Presentation, and Non-Technical Summary and small group meetings. Furthermore, an English advert ran in The New Times, a major national newspaper, on 8-December-2020; and e-mail invitations were sent from C-Quest Capital and a Rwandan based

implementation partner, ENEDOM, on 18-December-2020 and to 40 people from organizations around the country including government, NGOs, stove manufacturers, project developers, academia and private and public-sector entities involved with cookstoves and/or sustainable energy. The email included a link to the CQC website announcement where additional stakeholders from left feedback electronically.

A national stakeholder meeting was held in Kigali and attended in person by ten individuals, with another three additional stakeholders contributing feedback in an online chat. In summary, there was a considerable amount of feedback regarding the stove project, mostly positive. Many appreciated the projects use of leveraging carbon finance, but there was interest in why the focus was specifically on rural areas and emphasized the need for a clean cooking program to be implemented at a national level reaching all classes. There was a question on whether the high temperature the stove reaches (600 C) during cooking, would damage individuals' cooking pots? SNV asked what the best practices of this project were in other operational countries as policy in Rwanda discouraged the free distribution of stoves, however, it was eventually agreed upon that the beneficiary involvement in the construction of their stove was enough. On this note, they felt there should be a standard for brick making/brick-makers as not everyone has access to the appropriate clay. The two stove per household model was also scrutinized as stakeholders believed not all households would be able to utilize two stoves effectively. The issue was raised to have the option of providing one or two stoves, based on the household's ability to efficiently use them. Overall, the meeting achieved the intended objectives and contributions from stakeholders were highly appreciated. Many found the introduction of the VERRA carbon model in achieving their clean cooking goals to be very insightful as it offers a faster and more sustainable methodology for beneficiaries (and the environment) to directly benefit from carbon credit generation. Complementing this program with existing global initiatives would continue give it a competitive advantage and there was resounding agreement to begin a stove program in 2021 in Rwanda.

Grievance Redressal mechanism.

PP follows a proper Feedback and Grievance Redress Policy and Procedure (Being provided to the VVB for reference), which ensures that project-affected communities and individual stakeholders' grievances are properly prioritized and addressed. These measures are taken to enhance PP's accountability and transparency and to support the project initiatives that can help the communities to identify adverse effects to them, or their environment which had not previously been identified and mitigated by PP.



Flow of information/complaint/grievances from stakeholder to PP

The Grievance Policy and Procedure is accessible as a phone number of the PP/implementing partner office in Rwanda is available on the brochure provided to each household, which is the most accessible manner to provide feedback. PP policy and procedure outline the process to evaluate and respond to complaints; the procedure details when a serious level complaint should be communicated to PP country teams and when it would be best addressed by the implementing partner.

2.3 Environmental Impact

No negative environmental impacts have been identified from the project and environmental impact assessment (EIA) is not required for the project.

2.4 Public Comments

The project was open for public comments between 26-November-2020 to 26-December-2020. No public comments were received during this period.

2.5 AFOLU-Specific Safeguards

This section is not applicable as the project is a non-AFOLU project.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

1. VCS Methodology VMR0006- “Methodology for Installation of High Efficiency Firewood Cookstoves”. (Version 1.1⁵)
2. Tool 30 version 3.0 – “Calculation of the fraction of non-renewable biomass”.(Version 3.0)⁶
3. Standard – “Sampling and Surveys for CDM project activities and programmes of activities”(Version 09.0⁷)

3.2 Applicability of Methodology

Applicability criterion	How the project complies
Project activities must introduce efficiency improvements in thermal applications of non-renewable biomass in the following premises: households, community-based kitchens, institutions (e.g., schools) or small and medium sized enterprises (SMEs).	The proposed project involves deployment of Improved cookstoves which leads to improvement in thermal application of firewood (non-renewable biomass) in household premises..
The project stove shall have specified high-power thermal efficiency of at least 25% per the manufacturer’s specifications and shall exclusively use woody biomass and can be single pot or multi-pot; in case of project stove replacing fossil fuel baseline stove, it shall exclusively use renewable biomass.	For the present project activity instance, TLC Rocket Stoves (or other ICS models may be added) are envisaged to be distributed in end user households. TLC Rocket stoves having an efficiency of 34.5% as per the manufacture’s specification.
Non-renewable biomass has been used in the project region since 31 December 1989, using survey methods or referring to published literature, official reports or statistics;	Non-renewable biomass has been used since 31 December 1989 in Rwanda as demonstrated below the table.
For the specific case of biomass residues processed as a fuel (e.g. briquettes, wood chips) it shall be demonstrated that: (a) It is produced using exclusively renewable biomass (more than one type of biomass may be used). (b) The consumption of the fuel should be monitored during the crediting period and (c) Energy use for renewable biomass processing (e.g. shredding and compacting in the case of briquetting) may be considered as equivalent to the upstream	Not applicable for this project activity. The ICS is introduced as energy efficiency measure to replace baseline stoves and will reduce the use of non-renewable biomass used for cooking.

⁵ <https://verra.org/methodology/vmr0006-methodology-for-installation-of-high-efficiency-firewood-cookstoves/>

⁶ https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-30-v4.0.pdf/history_view

⁷ https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20210531160756223/Meth_Stan05.pdf

emissions associated with the processing of the displaced fossil fuel and hence disregarded.	
Both 'Projects' and 'Large Projects' can use this methodology.	The proposed project is expected to generate 3,393,437tonnes CO ₂ per year which is greater than 300,000 tonnes CO ₂ per year and thus qualifies as a "large project" as per section 3.9.1 of the VCS Standard version 4.3
The CDM-PDD or CDM-PoA-DD/CPA-DD shall explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g. programme logo).	Each ICS in this project will be identified by a unique combination of customer name and geographical location, as well as a serial number. The serial number will be a unique number which will allow for a clear distinction between the stoves. No individual serial number can be repeated within the project, thus ensuring that each stove is counted only once in the proposed project.
The CDM-PDD or CDM-PoA-DD/CPA-DD shall also explain how the proposed procedures prevent double counting of emission reductions, for example to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices.	The stove manufacturers/wholesale providers/end users shall sign an undertaking stating clearly that the PP or an entity authorized by it shall be the sole owner of the VCUs arising from the project

Evidence that the non-renewable biomass has been in use since 1989

Non-renewable biomass has been in use since 31-December-1989 as evidenced by various FAO statistical data. According to Global Forest Resources Assessment 2020⁸ (Page 18), forest area decreased from 317,000 ha in year 1990 to 276,000 ha in year 2020.

According to the National Strategy for Transformation (2017-2024)⁹, more than 83.3% of households in 2014 were dependent on firewood as a source of energy for cooking. Thus, it can be concluded that losses have been reported in forest areas from 1989 thereby hinting that fuelwood used across Rwanda comes from non-renewable sources.

⁸ <https://www.fao.org/3/ca9878fr/ca9878fr.pdf>

⁹ <https://www.undp.org/rwanda/rwanda%E2%80%99s-health-and-forests-need-clean-break-cook-stoves>

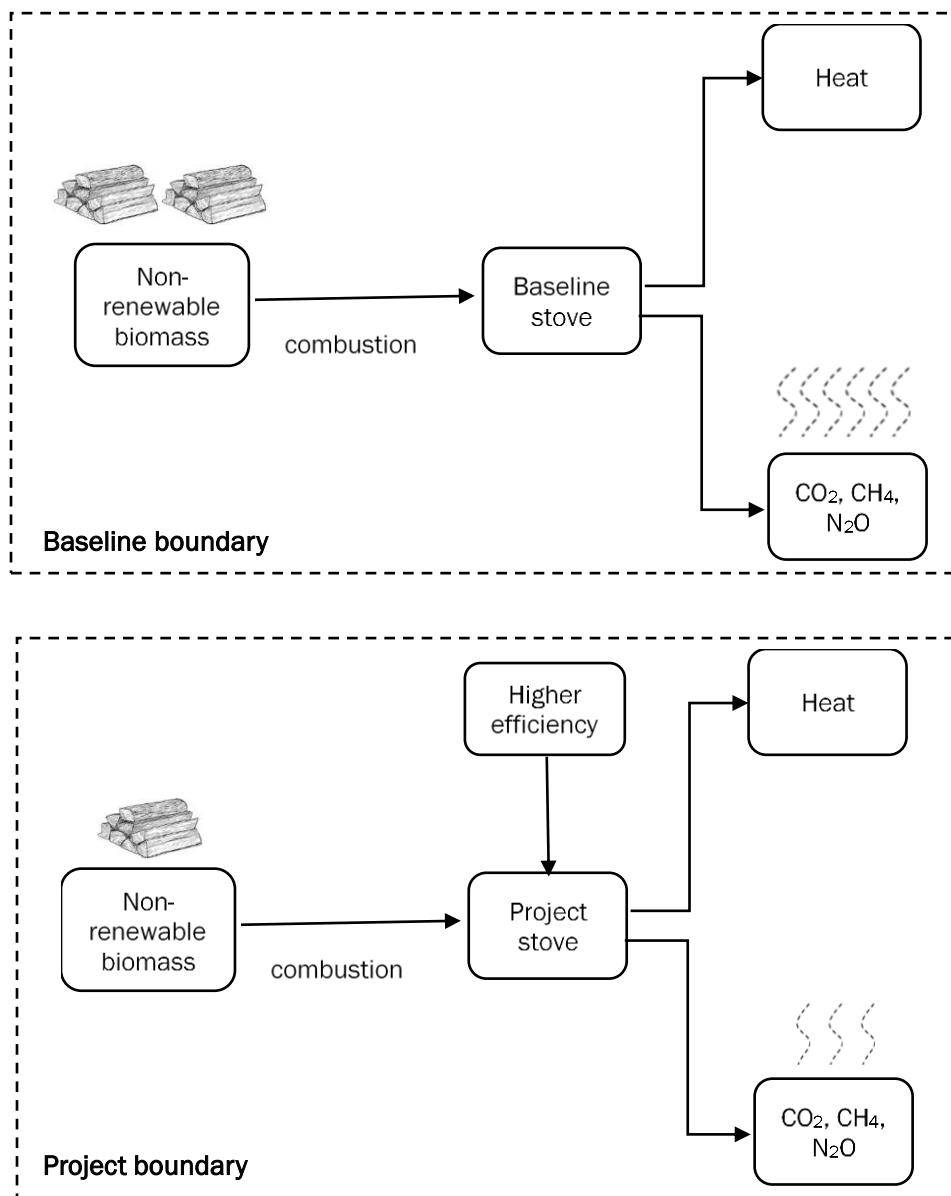
Catégorie	Type de forêts	Superficies en Ha							
		1960	1970	1980	1990	1996	1999	2001	2005
Forêts Naturelles	Nyungwe+Cyamudongo	114000	108800	97500	97500	94500	92400	92400	92400
	Gishwati	28000	28000	23000	8800	3800	600	600	1000
	Mukura	3000	3000	2100	2100	1600	1200	1200	1200
	Parc des volcans	34000	16000	15000	12760	12760	12000	12000	12000
	Parc de l'Akagera	241000	241000	241000	241000	241000	90000	90000	90000
	Domaine de chasse	64000	45000	45000	34000	0	0	0	0
	Forêts galeries de l'Est	150000	150000	90000	55000	30000	25000	25000	25000
Plantations forestières	Plantations forestières	24500	27160	80000	247500	232500	252000	306663	323400
TOTAL		658500	618960	593600	698660	616160	473200	527863	545000

Source : MINIRENA/Service Forestier National 2006.

3.3 Project Boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Emission from use of non-renewable biomass/Fossil fuel	CO ₂	Yes	Major source
		CH ₄	Yes	Major source
		N ₂ O	Yes	Major source
		Other	No	No other source identified
Project	Emission from use of non-renewable biomass	CO ₂	Yes	Major source
		CH ₄	Yes	Major source
		N ₂ O	Yes	Major source
		Other	No	No other source identified

A representation of the baseline boundary and project boundary are given as below.



3.4 Baseline Scenario

The baseline scenario is the continued use of three stone fired cookstoves or a conventional device with no improved combustion air supply or flue gas ventilation (that is without a grate or a chimney) by the target population by burning non-renewable woody biomass to meet the thermal energy needs. The project cookstoves (ICS) will be deployed substituting the less efficient baseline stoves.

The baseline cookstoves are three-stone fires or conventional systems, which lacks efficient combustion chambers/air supply mechanism and flue gas ventilation system, leading to more wood fuel use and higher GHG emissions. While the project stoves continue to use the same

wood fuel, with higher efficiency, it reduces the need (saves) to burn more wood fuel for the same purposes (household cooking). Baseline scenario can also be verified from the registration cum consent deed signed by the registered stove owner at the time of distribution of the project device (ICS), which confirms that, registered user voluntarily adopting the project stove and was using the three stone fire or conventional device (without a grate or a chimney) before adopting the project stove.

3.5 Additionality

The methodology uses activity method for the demonstration of additionality.

Activity Method

Step 1: Regulatory Surplus

There is no mandated government programme or policy in host country of this project ensuring the distribution of domestic fuel-efficient cookstoves. The project is not mandated by any law, statute or other regulatory framework, or for UNFCCC non-Annex I countries, any systematically enforced law, statute or other regulatory framework.

Households may only participate voluntarily in this project. It is hereby confirmed that the proposed project is a voluntary coordinated action by CQC.

Step 2: Positive List

As per Section 3.2, the project meets the applicability conditions of the methodology which represent the positive list.

The project installs the ICS at zero cost to the household and has no other source of revenue other than the sale of GHG credits.

The project is not implemented as part of government schemes or supported by multilateral funds.

Conclusion: As the project fulfills the conditions above, it is deemed additional.

Step 3: Project Method

For any project activity where stoves are not provided at zero cost to the end-user or has any other source of revenues other than the sale of GHG credits, then the project activity shall apply investment analysis method set out in the CDM Tool for the Demonstration and Assessment of Additionality included in AMS-II.G to determine that the proposed project activity is either: 1) not the most economically or financially attractive, or 2) not economically or financially feasible.

The project installs the ICS at zero cost to the household and has no other source of revenue other than the sale of GHG credits. So, Project method will not applicable here.

3.6 Methodology Deviations

The project did not apply any methodology deviations.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The methodology (VMR0006- Methodology for Installation of High Efficiency Firewood Cookstoves, Version 1.1) does not account for baseline emissions separately, but instead quantifies net emission reductions achieved by the project. Please refer to Section 4.4.

4.2 Project Emissions

The methodology VMR0006- Methodology for Installation of High Efficiency Firewood Cookstoves, Version 1.1) does not account for project emissions separately, but instead quantifies net emission reductions achieved by the project. Please refer to Section 4.4.

4.3 Leakage

Leakage shall be considered as default 0.95 in accordance with methodology (The methodology (VMR0006- “Methodology for Installation of High Efficiency Firewood Cookstoves”(Version 1.1)

4.4 Net GHG Emission Reductions and Removals

The improved cookstove is introduced as energy efficiency measure in the project, therefore equations 1 and 2 of the methodology will be applied to calculate the net GHG emission reductions.

$$ER_y = \sum_i \sum_j ER_{y,i,j} \quad \text{Equation (1)}$$

Where:

i	= Indices for the situation where more than one type/model of improved cookstove is introduced to replace three-stone fire
j	= Indices for the situation where there is more than one batch of improved cookstove of type i
ER_y	= Emission reductions during year y in t CO ₂ e
$ER_{y,i,j}$	= Emission reductions by improved cookstove of type i and batch j during year y in t CO ₂ e

$$ER_{y,i,j} = B_{y,savings,i,j} \times NCV_{wood\ fuel} \times f_{NRB,y} \times (EF_{wf,CO_2} + EF_{wf,non\ CO_2}) \times N_{y,i,j} \times 0.95 \quad \text{Equation (2)}$$

Where:

$B_{y,savings,i,j}$	=	Quantity of woody biomass that is saved in tonnes per improved cookstove of type i and batch j during year y
$f_{NRB,y}$	=	Fraction of woody biomass that can be established as non-renewable biomass (fNRB) ¹⁰
$NCV_{wood\ fuel}$	=	Net calorific value of the non-renewable woody biomass that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/tonne) ¹¹
EF_{wf,CO_2}	=	CO ₂ emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 112 tCO ₂ /TJ) ¹²
$EF_{wf,non\ CO_2}$	=	Non-CO ₂ emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 26.23 tCO ₂ /TJ) ¹³
$N_{y,i,j}$	=	Number of improved cookstoves of type i and batch j operating during year y
0.95	=	Discount factor to account for leakage

The quantify of woody biomass saved due to implementation of improved cookstoves to be estimated using equation below:

$$B_{y,savings,i,j} = B_{y=1,new,i,survey} \times \left(\frac{\eta_{new,y,i,j}}{\eta_{old}} - 1 \right) \quad \text{Equation (4)}$$

where

η_{old}	=	Efficiency of baseline cookstove
$\eta_{new,y,i,j}$	=	Efficiency of the improved cookstove type i and batch j determined through water boiling test (WBT) during year y Alternatively, efficiency may be determined using Equation 5.
$B_{y=1,new,i,j,survey}$	=	Annual quantity of woody biomass used by improved cookstoves in tonnes per device of type i and batch j , determined in the first year of the implementation of the project through a sample survey.

¹⁰ Default values endorsed by designated national authorities and approved by the Board are available at <https://cdm.unfccc.int/DNA/fNRB/index.html>

¹¹ 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 1 Introduction

¹² 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion

¹³ 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion

$$\eta_{new,y,i,j} = \eta_p \times (DF_n)^{y-1} \times 0.94 \quad \text{Equation (5)}$$

where

- η_p = Efficiency of project stove (fraction) at the start of project activity.
- $(DF_n)^{y-1}$ = Discount factor to account for efficiency loss of project cookstove per year of operation (fraction). This value may be based on actual monitoring or based on manufacturer's declaration on expected loss in efficiency or through publicly available literature on relevant industry standards. Alternatively default value of 0.99 efficiency loss per year can be considered.
- 0.94 = Adjustment factor to account for uncertainty related to project cookstove efficiency test.

For ex-ante calculation purpose, the assumption below is applied.

1. The project will install up to 500,000 ICS.
2. ICS installation to be implemented in 4 years wherein 125,000 ICS will be installed in each year.
3. The life span of ICS is 10 years.
4. $B_{y=1,new,i,j,survey}$ is assumed as 2.3543 tonnes / device / year.

The value for $B_{y=1,new,i,j,survey}$ has been calculated based on baseline wood fuel consumption values derived considering literature review (PP considering the literature review of Burundi, which is a neighboring country, as the related study data of Rwanda is not available) and considering average household size of 4.3 for the purpose of ex ante calculations of estimated net emission reductions. Actual value of $B_{y=1,new,i,j,survey}$ will be identified at the time of first verification through monitoring survey.

Parameter	value	unit	source	
firewood consumption/Person	3.00	kg/day	<u>Toward policies for climate change mitigation: "Barriers for family-sized biogas in the District of Gihanga, Burundi" (wiley.com)</u>	Page no.2,para 1
Household size of Rwanda(rural)	4.3		<u>https://dhsprogram.com/pubs/pdf/FR316/FR316.pdf</u>	Page no.15, section 2.2(mean size of rural HH between 2014-2015)

firewood consumption/household	12.90	kg/day	calculated	
firewood consumption/household/year	4.7085	tonnes/household/year	calculated	
firewood consumption/device/year	2.3543	tonnes/device/year	calculated	

5. Annual stove loss rate is estimated at 5%.

6. Fraction of non-renewable biomass (fNRB) calculation for Rwanda

Variable	Value	Source
Total Woody biomass consumption (t/yr) (H)	16,742,127	UN Data ¹⁴ , world bank data ¹⁵ FAO(in accordance with section 4.0 of the TOOL30, v03.0).
Renewable biomass (t/yr) (RB)	382,778	2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Local study
Non-renewable biomass (t/yr)	16,359,348	Calculated (Equation 2 of CDM tool 30, v03.0)
Fraction of non-renewable biomass	0.97	Calculated (Equation 1 of CDM Tool 30, v03.0)

The difference between woody biomass consumption and renewable biomass is considered to be non-renewable (CDM Tool 30, Equation 2). Non-renewable biomass utilization in Rwanda is, therefore, estimated to be 16,359,348t/yr. The fraction of non-renewable biomass is the quotient of the non-renewable and the total biomass (CDM Tool 30, Equation 1). The fraction of non-renewable biomass for Rwanda is, therefore, estimated to be 0.97.

C4 Ecosolution (third party to study and derive the fNRB value for Rwanda) has used the latest historical data available in the UN Data and world bank data (for 2021).

Determination of number of ICS operating during year y

$N_{(y,i,j)}$ = Number of project devices operational in previous year x [1 -annual stove loss rate(%)]
+Project Devices installed in current year 'y'

Example of calculation:

¹⁴ UNdata | record view | Fuelwood

¹⁵ Urban population - Rwanda | Data (worldbank.org), Rural population - Rwanda | Data (worldbank.org)

If $y = 2$,

$$N_{y,i,j} = 118,750 \times [1-5\%] + 125,000$$

$$= 118,000 \times 95\% + 125,000$$

$$= 237,813$$

Hence, the number of ICS operating during year y is as below:

Year (y)	$N_{y,i,j}$
09-October-2021 to 08-October-2022	118,750
09-October-2022 to 08-October-2023	237,813
09-October-2023 to 08-October-2024	350,922
09-October-2024 to 08-October-2025	458,376
09-October-2025 to 08-October-2026	435,457
09-October-2026 to 08-October-2027	413,684
09-October-2027 to 08-October-2028	393,000
09-October-2028 to 08-October-2029	373,350
09-October-2029 to 08-October-2030	354,682
09-October-2030 to 08-October-2031	336,948

Determination of efficiency of ICS during year y

$$\eta_{new,y,i,j} = \eta_p \times (DF_n)^{y-1} \times 0.94$$

Where

$$\eta_p = 34.5\%$$

$$DF_n = 0.99$$

Example of calculation:

If $y = 2$

$$\eta_{new,y,i,j} = 34.5\% \times (0.99)^{2-1} \times 0.94$$

$$= 32.11\%$$

Hence the efficiency of ICS during year y is as below:

Year (y)	$\eta_{new,y,i,j}$
----------	--------------------

09-October-2021 to 08-October-2022	32.43%
09-October-2022 to 08-October-2023	32.11%
09-October-2023 to 08-October-2024	31.78%
09-October-2024 to 08-October-2025	31.47%
09-October-2025 to 08-October-2026	31.15%
09-October-2026 to 08-October-2027	30.84%
09-October-2027 to 08-October-2028	30.53%
09-October-2028 to 08-October-2029	30.23%
09-October-2029 to 08-October-2030	29.92%
09-October-2030 to 08-October-2031	29.63%

Determination of quantity of woody biomass that is saved in tonnes per ICS during year y

$$B_{y,savings,i,j} = B_{y=1,new,i,survey} \times \left(\frac{\eta_{new,y,i,j}}{\eta_{old}} - 1 \right)$$

Example of calculation:

If y= 2,

$$B_{y,savings,i,j} = 2.3543 \times [(0.3211/0.1) - 1]$$

$$= 5.2042$$

Year (y)	$B_{y=1,new,i,survey}$ (tonnes/device/ year)	$\eta_{new,y,i,j}$ (tCO ₂ e/TJ)	η_{old} (tCO ₂ e/ TJ)	$B_{y,savings,i,j}$ (tonnes/device/year)
09-October-2021 to 08-October-2022	2.3543	32.43%	0.1	5.2806
09-October-2022 to 08-October-2023	2.3543	32.11%	0.1	5.2042
09-October-2023 to 08-October-2024	2.3543	31.78%	0.1	5.1286
09-October-2024 to 08-October-2025	2.3543	31.47%	0.1	5.0538

09-October-2025 to 08-October-2026	2.3543	31.15%	0.1	4.9797
09-October-2026 to 08-October-2027	2.3543	30.84%	0.1	4.9064
09-October-2027 to 08-October-2028	2.3543	30.53%	0.1	4.8338
09-October-2028 to 08-October-2029	2.3543	30.23%	0.1	4.7619
09-October-2029 to 08-October-2030	2.3543	29.92%	0.1	4.6908
09-October-2030 to 08-October-2031	2.3543	29.63%	0.1	4.6203

Determination of emission reductions by ICS during year y

$$ER_{y,i,j} = B_{y,savings,i,j} \times NCV_{wood\ fuel} \times f_{NRB,y} \times (EF_{wf,CO_2} + EF_{wf,non\ CO_2}) \times N_{y,i,j} \times 0.95$$

Where

$$NCV_{wood\ fuel} = 0.0156 \text{ TJ/tonne}$$

$$f_{NRB,y} = 0.97$$

$$EF_{wf,CO_2} + EF_{wf,non\ CO_2} = 112 + 26.23 = 138.23 \text{ tCO}_2/\text{TJ}$$

Example of calculation:

If y=2,

$$ER_{y,i,j} = 5.2042 \times 0.0156 \times 0.97 \times 138.23 \times 237,813 \times 0.95$$

$$= 2,459,313 \text{ tCO}_2$$

Year	$B_{y,savings,i,j}$ (tonnes/de vice/year)	$NCV_{wood\ fuel}$ (Tj/tonne)	$f_{NRB,y}$ (Fraction)	EF_{wf,CO_2} + $EF_{wf,non\ CO_2}$ (tCO ₂ e/TJ)	$N_{y,i,j}$ (Units)	$ER_{y,i,j}$ (tCO ₂ e/ann um)
09-October-2021 to 08-October-2022	5.2806	0.0156	0.97	138.23	118,750	1,246,056

09-October-2022 to 08-October-2023	5.2042	0.0156	0.97	138.23	237,813	2,459,313
09-October-2023 to 08-October-2024	5.1286	0.0156	0.97	138.23	350,922	3,576,315
09-October-2024 to 08-October-2025	5.0538	0.0156	0.97	138.23	458,376	4,603,241
09-October-2025 to 08-October-2026	4.9797	0.0156	0.97	138.23	435,457	4,308,977
09-October-2026 to 08-October-2027	4.9064	0.0156	0.97	138.23	413,684	4,033,240
09-October-2027 to 08-October-2028	4.8338	0.0156	0.97	138.23	393,000	3,774,877
09-October-2028 to 08-October-2029	4.7619	0.0156	0.97	138.23	373,350	3,532,806
09-October-2029 to 08-October-2030	4.6908	0.0156	0.97	138.23	354,682	3,306,011
09-October-2030 to 08-October-2031	4.6203	0.0156	0.97	138.23	336,948	3,093,541

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
Year 2021-22 (09-October-2021 to 08-October-2022)	1,246,056	0	0	1,246,056
Year 2022-23 (09-October-2022 to 08-October-2023)	2,459,313	0	0	2,459,313
Year 2023-24 (09-October-2023 to	3,576,315	0	0	3,576,315

08-October-2024)				
Year 2024-25 (09-October-2024 to 08-October-2025)	4,603,241	0	0	4,603,241
Year 2025-26 (09-October-2025 to 08-October-2026)	4,308,977	0	0	4,308,977
Year 2026-27 (09-October-2026 to 08-October-2027)	4,033,240	0	0	4,033,240
Year 2027-28 (09-October-2027 to 08-October-2028)	3,774,877	0	0	3,774,877
Year 2028-29 (09-October-2028 to 08-October-2029)	3,532,806	0	0	3,532,806
Year 2029-30 (09-October-2029 to 08-October-2030)	3,306,011	0	0	3,306,011
Year 2030-31 (09-October-2030 to 08-October-2031)	3,093,541	0	0	3,093,541
Total	33,934,377	0	0	33,934,377

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter

$f_{NRB,y}$

Data unit	Fraction
Description	Fraction of woody biomass saved by the project activity during year y that can be established as non-renewable biomass
Source of data	IPCC 2019 refinement, UN Data & FAO reports
Value applied	0.97
Justification of choice of data or description of measurement methods and procedures applied	This parameter shall be determined ex-ante. C4 EcoSolutions (Pty) Ltd was appointed as third party to study and derive the f_{NRB} value for Rwanda. The value has been calculated in accordance with Tool 30 version 3; using latest available versions of Food and Agriculture Organization (FAO) report, UN Data and other publicly available data that have been published by reliable sources.
Purpose of Data	Calculation of emission reductions
Comments	The report and calculation spreadsheet for f_{NRB} will be made available to VVB during the validation.

Data / Parameter	$NCV_{wood\ fuel}$
Data unit	TJ/tonne
Description	Net calorific value of the non-renewable woody biomass that is substituted or reduced
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 1 Introduction
Value applied	0.0156
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value
Purpose of Data	Calculation of emission reductions
Comments	No comments

Data / Parameter	EF_{wf,CO_2}
Data unit	tCO ₂ /TJ
Description	CO ₂ emission factor for the use of wood fuel in baseline scenario

Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion
Value applied	112
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value
Purpose of Data	Calculation of emission reductions
Comments	No comments

Data / Parameter	$EF_{wf,non\ CO2}$
Data unit	tCO ₂ /TJ
Description	Non-CO ₂ emission factor for the use of wood fuel in baseline scenario
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion
Value applied	26.23
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value
Purpose of Data	Calculation of emission reductions
Comments	No Comments

Data / Parameter	η_{old}
Data unit	Fraction
Description	Efficiency of baseline cookstove
Source of data	Methodology default value
Value applied	0.1
Justification of choice of data or description of	A default value of 0.1 shall be used if baseline device is a three-stone fire using firewood (not charcoal), or a conventional device

measurement methods and procedures applied	with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney
Purpose of Data	Calculation of emission reductions
Comments	Baseline stove use can be confirmed from the registration form cum consent deed signed by each Household/stove owner at the time of installation of ICS which states that the “household does not currently own an improved cookstove and is using the improved cookstove voluntarily to replace a traditional three stone fire or pot support.

Data / Parameter	η_p
Data unit	Fraction
Description	Efficiency of project stove at the start of project activity.
Source of data	Stove efficiency tests performed by Aprovecho Research centre.
Value applied	0.345
Justification of choice of data or description of measurement methods and procedures applied	This parameter shall be determined ex-ante
Purpose of Data	Calculation of $\eta_{new,y,i,j}$
Comments	No Comments

5.2 Data and Parameters Monitored

Data / Parameter	$N_{y,i,j}$
Data unit	Number
Description	Number of project devices of type i and batch j operating during year y
Source of data	Monitoring Survey
Description of measurement methods and procedures to be applied	Measured directly or based on a representative sample. Sampling standard shall be used for determining the sample size to achieve 90/10 confidence precision according to the latest

	version(version9.0) of Standard for sampling and surveys for CDM project activities and programme of activities.
Frequency of monitoring/recording	At least once in every two years
Value applied	For ex-ante emission reduction calculation, it is assumed that the project will distribute up to 500,000 ICS years (125,000 ICS will be installed each year). Actual number of operating stoves will be identify during each periodic verification.
Monitoring equipment	Monitoring survey
QA/QC procedures to be applied	The unique reference number of each stove shall be logged in the monitoring database showing the total number of stoves. Data from the sampling plan will be collected in each monitoring period by trained project staff and applied in the emission reduction calculations. Internal cross-checks by the Project Proponent will be undertaken as Quality Check.
Purpose of data	Calculation of emission reductions
Calculation method	Proportion of operational stoves obtained from the survey is multiplied by the total commissioned stoves to arrive at this value
Comments	As per applied methodology, PP needs to conduct the monitoring survey at least once in every 2 years (biennial), however PP is planning to conduct monitoring survey annually (at least once every year).

Data / Parameter	$\eta_{new,i,j}$
Data unit	Fraction
Description	Efficiency of the device of each type i and batch j implemented as part of the project activity.
Source of data	Calculation
Description of measurement methods and procedures to be applied	To adopt Option V given in the methodology: “Efficiency of the improved cookstoves to be estimated using equation 5 above where loss in efficiency per year is calculated, and therefore this parameter does not need to be monitored”
Frequency of monitoring/recording	Annually

Value applied	For ex-ante calculation, the value below is applied.																						
	<table> <tr> <th>Year (y)</th><th>$\eta_{new,y,i,j}$</th></tr> <tr><td>1</td><td>32.43%</td></tr> <tr><td>2</td><td>32.11%</td></tr> <tr><td>3</td><td>31.78%</td></tr> <tr><td>4</td><td>31.47%</td></tr> <tr><td>5</td><td>31.15%</td></tr> <tr><td>6</td><td>30.84%</td></tr> <tr><td>7</td><td>30.53%</td></tr> <tr><td>8</td><td>30.23%</td></tr> <tr><td>9</td><td>29.92%</td></tr> <tr><td>10</td><td>29.63%</td></tr> </table>	Year (y)	$\eta_{new,y,i,j}$	1	32.43%	2	32.11%	3	31.78%	4	31.47%	5	31.15%	6	30.84%	7	30.53%	8	30.23%	9	29.92%	10	29.63%
Year (y)	$\eta_{new,y,i,j}$																						
1	32.43%																						
2	32.11%																						
3	31.78%																						
4	31.47%																						
5	31.15%																						
6	30.84%																						
7	30.53%																						
8	30.23%																						
9	29.92%																						
10	29.63%																						
Monitoring equipment	Not Applicable																						
QA/QC procedures to be applied	Not Applicable																						
Purpose of data	Calculation of emission reductions																						
Calculation method	Calculation to be performed using equation below: $\eta_{new,y,i,j} = \eta_p \times (DF_n)^{y-1} \times 0.94$																						
Comments	No Comments																						

Data / Parameter	$B_{y=1,new,i,j,survey}$
Data unit	Tonnes
Description	Annual quantity of woody biomass used by project devices in tonnes per device of type i and batch j, determined in the first year of the implementation of the project through a sample survey
Source of data	Monitoring survey

Description of measurement methods and procedures to be applied	Minimum sample size of each type i and batch j should be in line with the latest version of Standard for sampling and surveys for CDM project activities and programme of activities or guidelines provided in methodology Section 8.4 option (b). Determined in the first year of the introduction of the devices (e.g. during the first year of the crediting period, $y=1$) through measurement campaigns at representative households and/or sample survey. Sample surveys to estimate this parameter, that are solely based on questionnaires or interviews (i.e. that do not implement measurement campaigns) may only be used if the following conditions are satisfied. (i) Baseline cookstoves have been completely decommissioned and only improved cookstoves are exclusively used in the project households. (ii) If multiple devices are used in the project, it is possible from the results of the survey questions to clearly differentiate the quantity of firewood being used by each device. In other words, if more than one device, or another device that consumes firewood, are in use in project households, then the sample survey needs to distinguish the quantity of firewood used by the project device and the other devices that use firewood.
Frequency of monitoring/recording	Will be determined in the first year of project implementation
Value applied	For ex-ante calculation, the value is assumed as 2.3543tonnes/device/year. The value for $B_{y=1,new,i,j,survey}$ has been calculated based on baseline wood fuel consumption values derived considering <u>literature review</u> (PP considering the literature review of Burundi, which is a neighboring country, as the related study data of Rwanda is not available) and considering <u>average household</u> size of 4.3 for the purpose of ex ante calculations of estimated net emission reductions.
Monitoring equipment	Weighing scale will be used to measure the wood consumption in project stoves
QA/QC procedures to be applied	Survey to be conducted as per requirements of the methodology. Monitoring equipment will be calibrated as per the manufacturer 's guidelines
Purpose of data	Calculation of emission reductions
Calculation method	Mean woody biomass consumption of sample stoves to be applied across entire population.
Comments	Actual value of $B_{y=1,new,i,j,survey}$ will be determined in the first year of the implementation of the project through a sample survey.

Data / Parameter	Life Span
Data unit	Number of years
Description	The operating lifetime of the project device. The life span should be reported if the methodology equation 5 is adopted to determine the project stove efficiency
Source of data	Manufacturer's specification
Description of measurement methods and procedures to be applied	TLC cookstoves will be manufactured to match the fixed design specification. This will be achieved using brick molds of specified dimensions to make bricks to be used for stove construction locally. This will ensure that each stove that is built at individual end user household measures exactly same as the dimensions specified by the manufacturer. Post construction, training will be provided to end users on use, care, and upkeep of these stoves. PP will conduct periodic audits and surveillance of the stoves that will be distributed under the project activity to ensure their proper functioning throughout the project lifetime.
Frequency of monitoring/recording	Once at the time of project stove installation
Value applied	10
Monitoring equipment	No equipment is required to monitor this parameter
QA/QC procedures to be applied	This parameter will be referred from the Manufacturer's specification
Purpose of data	Calculation of emission reductions
Calculation method	Not Applicable
Comments	No Comments

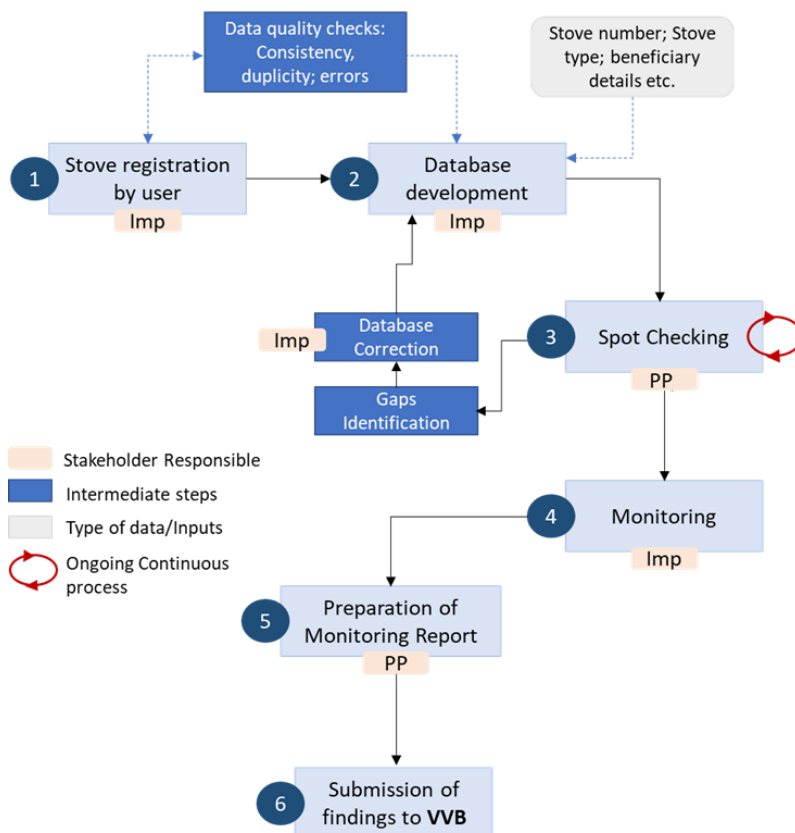
5.3 Monitoring Plan

The Monitoring Plan applied in this project involves several key elements that ensure that the PP and/or project Implementer have high-quality, unbiased, and reliable information regarding the performance of the project in terms of implementation and outcomes, and for the purposes of calculating Verified Carbon Units (VCUs) following VMR 0006 version 1.1 on the basis of the amount of non-renewable biomass saved by the ICS in the project activity. The key elements are the following:

- Data collection procedures
- Distribution and Monitoring Database
- Spot Checking of ICS (ongoing)

- Sample Plan for the Monitoring Survey
- Data Quality, Consistency and Duplication Checks
- Monitoring Reporting

The below flow-chart illustrates the roles and responsibilities of the parties during the implementation of the monitoring plan for the project. In the below flowchart, the project implementer is abbreviated to “Imp” and can be the PP or another party authorized by the PP.



Below is the description of the above steps on the flow-chart.

- Implementor : User registers stove:** Project implementer will collect/receive the necessary information requested in the Registration process from the user. Means of collecting this information may be through a physical registration card filled by project Implementor staff, retailers, end-users, or partner organization’s staff, or through the use of ICTs or SMS. Project Implementers’ staff shall double check the accuracy of information provided, and request for field staff additional clarifications if needed.
- Implementor : Data logged into database:** Project implementer’s trained staff will input the data in the database either manually (if data collected from physical registration card) or this will be automatically input if data was collected using ICTs or SMS. Project implementer staff shall double check the information included on the database and check for duplications. Any duplicate information shall be investigated, and errors corrected or excluded from the database if it is a true duplicate entry.

3. **Project Proponent : Spot- checking (ongoing):** Project Proponent has a procedure for internal auditing called spot-checking, where field staff randomly selects households to ensure proper functioning of the project stoves throughout the project lifetime and to ensure that the project stoves continue to work as efficient as specified by the manufacturer. Field staff of project proponent randomly select units included in the database and visit or contact the stove owners to cross-check the information on the database with the factual evidence in the field. Any inconsistencies found (e.g., change in the address of a user) gets updated on the database, and in case, ICS are found to be no longer in use, they get clearly marked as such and is excluded from emission reductions calculations.

During the spot check, if any of the project stoves was found to be in poor working condition like cracked/damaged or with missing metal parts, the stove owner repairs the stove as they have been trained during installation. If the stove owners are unsuccessful in repairing, the field staff informs the implementer to provide the necessary assistance and the missing metal parts are reported to the management of implementer who then either provides the parts from their existing stocks or request the same to project proponent who procured those from their suppliers. Implementing partner then provides those parts to the stove owners. This is a continuous process.

4. **Implementor : Monitoring :** Project implementer will follow the requirements to collect the necessary information for a monitoring report. The project proponent will assess all monitoring data and produce a monitoring report for the VVB to verify corresponding to the preceding monitoring period. This report will present the data relating to the emission reductions generated by those batches during the monitoring period.
5. **Project Proponent: Preparation of monitoring report:** The Project Proponent will prepare the final monitoring report to be provided to the VVB for verification of emission reductions. A copy of the monitoring report will remain with the Project Proponent.

The PP will coordinate and manage each project Implementer and assist them in implementing each element of the monitoring plan. The monitoring plan shall be elaborated in accordance with the Sampling Plan below. Also, regular audits conducted throughout the project lifetime ensures proper functioning of stoves and determination of the lifespan of the same.

Sampling Plan

As per Standard for Sampling and Surveys for CDM Project Activities and Programme of Activities; version 09.0, the sampling plan is the following:

(a) Sampling design

Due to the large number of ICS envisioned to be distributed under the project, it is not economically feasible to monitor each individual ICS unit distributed. Therefore, representative sampling shall be undertaken for each batch of ICS.

Samples will be drawn from ICS recorded in the project database administered by the project proponent.

I. Objectives and Reliability Requirements

The objective is to obtain an unbiased and reliable estimate of the proportion or mean value of the following key variables over the course of the crediting period.

Monitored Parameters:

The project involves introduction of improved cook stoves as energy efficiency measure

No.	Monitoring parameters	Sampling parameters	Parameter type	Monitoring frequency
1	$N_{y,i,j}$ Number of project devices of type i and batch j operating during year y	Proportion of ICS still in operation	Proportion	At least once in two years
2	$B_{y=1,new,i,j,survey}$ Quantity of woody biomass used by project devices in tonnes per device of type i . and batch j	Daily consumption of woody biomass per ICS	Mean value	Determined in the first year of project implementation

(ii) Target Populations:

The target population will be the complete set of appliances (ICS) deployed under the project.

(iii) Sampling Frame

It is expected that the geographical locations do not have influence on the parameters of interest that is $N_{y,i,j}$, & $B_{y=1,new,i,j,survey}$. Therefore, these parameters can be assumed to be highly homogeneous regardless of how the end user group and distribution/installation location is defined since the end users shall always be households and geographical location is restricted to the project boundary. The Sampling Frame for individual parameters have been defined in the following paragraphs.

The sample frame refers to all the information sources on the Database. The SMS/ICT system is the primary mechanism for data collection for newly distributed ICS and the Monitoring Survey (which includes a household questionnaire and visual inspection of ICSs) that will be used throughout the lifetime of the project. The SMS/ICT data is used to populate the stoves Database and the Monitoring Survey follows the “Standard for Sampling and Surveys for CDM Project Activities and Programme of Activities”; version 09.

All the TLC rocket stoves distributed under this project shall be grouped together to create a Primary Sampling Unit which is homogenous. The procedure to determine the sample of households will ensure that they adequately represent the broader project population, minimizing sampling error. Using, a 90 per cent confidence level, and a 10 per cent margin of error, a random sample will be selected from the target population.

1) Sampling frame for proportion of ICS still in operation ($N_{y,i}$)

The first step is to identify the Primary Sampling Units.

Factors such as model of the ICS do not effect parameter $N_{y,j}$. Therefore Primary sampling unit in case of parameters $N_{y,j}$ is independent of model of stove.

2) Quantity of woody biomass used by project devices in tonnes per device ($B_{y=1,new,i,survey}$)

The Parameter $B_{y=1,new,i,survey}$ shall vary in accordance with its model. Hence for this parameter, the Primary Sampling Unit shall be defined as the group of ICSs of the same model. If two different ICS models being implemented in the same project – this will form two Primary Sampling Units.

The value will be determined in the first year of the introduction of the devices through the monitoring survey.

Under this project two TLC-CQC Rocket Stoves will be installed in each household, which are classified as Project stove 1 and project stove 2. At the time of monitoring survey, field staff record the same in survey forms and in the spreadsheet. Therefore, firewood consumed for each project stove and baseline stove can be distinguished clearly.

Calibration of weighing scales use for measuring the fuel wood will be done in household before start using on site.

(iv) Sampling Method

The sampling method for the monitored parameters $N_{y,i,j}$, and $B_{y=1,new,i,survey}$ is Simple Random Sampling and samples will be randomly selected from the primary sampling units as illustrated above. To ensure a random selection of ICS, random number generators shall be applied. Each ICS in the target population is uniquely identifiable by its unique ID number. Each ICS can thus be allocated a Sample Selection Number in each monitoring period, starting at 1 and increasing up to the total number of ICS in the Database for that pre-defined sampling frame. Applying the random number generators, the ICS can then be randomly chosen from the defined population up to the required sample size as calculated by the project proponent.

To determine the parameters, sampling will involve the following approaches (outcome in brackets):

$N_{y,i,j}$	Visual inspection of the premises to see if ICS is operational and in use. Interview with end user if required to verify that ICS is still in use (Yes/No)
$B_{y=1,new,i,j,survey}$	Quantity of woody biomass used by project devices in tonnes per device (First year of installation).

(v) Sample Size

For the estimation of the proportion or mean value of the parameters investigated, the minimum sample size for each sample frame has to achieve the 90/10 confidence/precision.

The procedure to determine the sample of households will ensure that they adequately represent the broader project population, minimizing sampling error. Using a 90 per cent confidence level, and a 10 per cent margin of error, random samples will be selected from each Primary Sampling

Unit. The parameter number of stoves still in operation during the monitoring period as determined by the monitoring survey ($N_{y,j}$) will be estimated through sampling annually/biennially and the parameter Quantity of woody biomass used by project devices in tonnes per device $B_{y=1,new,i, survey}$ shall be estimated through sampling within the first year of project installation. $B_{y=1,new,i, survey}$ can be coupled with $N_{y,j}$ for the first year. Of the two parameters to be monitored, $N_{y,i,j}$ is proportion/percentage parameter and $B_{y=1,new,i, survey}$ is a mean value parameter.

Values for the proportions, mean values, and standard deviations are needed in order to compute the necessary sample size estimations. As per Guideline for Sampling and surveys for CDM project activities and programmes of activities, version 04.0, there are different ways available to obtain the estimates of the parameter of interest:

- (a) Refer to the result of previous studies and use these results.
- (b) In a situation where information from previous studies is not available, a preliminary sample as a pilot could be conducted and use that sample is used to provide the estimates;
- (c) Use best guesses based on the researcher's own experiences.

For the registration purpose of project, option (c) as stated above shall be applied. For the following monitoring periods, the estimates can be adjusted taken the results of the previous monitoring period(s) into account or the result from recent pilot study which is conducted after the previous monitoring periods.

Parameter $N_{y,i,j}$:

To estimate the sample size for the parameter $N_{y,i,j}$ the following equation is used:

$$n \geq \frac{1.645^2 N \times p \times (1 - p)}{(N - 1) \times 0.1^2 \times p^2 + 1.645^2 p \times (1 - p)}$$

Where:

n = Sample size

N = Population size (Total number of households/ICS)

p = Expected proportion

1.645 = Represents the 90% confidence required

0.1 = Represents the 10% relative precision

The following assumptions are made to exemplify the sample size calculation for the parameters.

1. An overview of the estimated sample sizes for a hypothetical population of 100,000 ICS units applying a level of 90/10 is provided below. It is likely that all the sample frames for each parameter will include fewer than 100,000 ICS in the first monitoring period, so this is a conservative approach. Hence, population size, N , is taken as 100,000 households/ICS (Assuming one ICS for one household).

2. It is expected at least 80% of ICS still in operation, hence the expected proportion p is taken as 0.8.

Sample size calculation:

The calculation of the required sample size for each parameter in the first monitoring period is illustrated below. In all cases a conservative approach is taken, however if for any parameter the required confidence/precision is not met then the PP will randomly select an additional sample and collect further data from this sample to ensure the pooled data meet or exceed the required thresholds.

Based on the above assumptions, the resulting sampling size is calculated as:

$$n \geq \frac{1.645^2 \times 100,000 \times 0.8(1 - 0.8)}{(100,000 - 1) \times 0.1^2 \times 0.8^2 + 1.645^2 \times 0.8(1 - 0.8)} = 67.61$$

Therefore, in this case a sample size of 68 is to be sampled from each primary sampling unit.

Parameter $B_{y=1, \text{new}, i, \text{survey}}$:

To estimate the sample size for parameter $B_{y=1, \text{new}, i, \text{survey}}$ the following equation is used:

$$n \geq \frac{1.645^2 NV}{(N - 1) \times 0.1^2 + 1.645^2 V}$$

Where:

$$V = \left(\frac{SD}{mean} \right)^2$$

n = Sample size

N = Population size (Total number of households/ICS)

mean = Expected mean

SD = Expected standard deviation

1.645 = Represents the 90% confidence required

0.1 = Represents the 10% relative precision

Based on the above assumptions, the sample size calculation would be

$$n \geq \frac{1.645^2 \times 100,000 \times \left(\frac{0.076}{0.38}\right)^2}{(100,000 - 1) \times 0.1^2 + 1.645^2 \times \left(\frac{0.076}{0.38}\right)^2} = 10.82$$

The resulting sample size based on the above equation is smaller than 30, since $B_{y=1, \text{new}, i, \text{survey}}$ is a numeric mean value (i.e. not a proportion or percentage) the Student's t-distribution shall be used as per paragraph 14 of "Standard for Sampling and Surveys for CDM Project Activities and Programme of Activities, version 09.0.

The sample size for parameter $B_{y=1, \text{new}, i, \text{survey}}$ under t-distribution is referred to the equation below:

$$n = \left(\frac{t_{n-1} \times SD}{0.1 \times \text{mean}} \right)^2$$

Where t_{n-1} is the value of the t-distribution for 90% confidence when the sample size is n. Since the sample size is not known yet, the first step is to use the value for 90% confidence when the sample is large, i.e., 1.645 and then redefine the calculation.

$$n = \left(\frac{1.645 \times 0.076}{0.1 \times 0.38} \right)^2 = 10.82$$

Thus n is rounded up to 11.

The calculation now needs to repeat using t-value for 90% confidence and n = 11

$$n = \left(\frac{2.228 \times 0.076}{0.1 \times 0.38} \right)^2 = 19.86$$

And n is rounded to 20.

The calculation now need to repeat using t_{n-1} value for n = 20. The process should be iterated until there is no change to the value of n.

t_{19-1}	2.093
n=	17.52
Round up	18

t_{18-1}	2.110
n=	17.81

Round up	18
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The repeated calculation shows that $n = 18$. Thus, the sample size to be sampled from each sampling unit is 18.

Since parameters $N_{y,j,j}$, and $B_{y=1,new,i, survey}$ share the same sampling units, PP may choose to have one common survey for these two parameters with largest number of sample size between these two parameters being chosen. Sampling more than one parameter within the same sample (household) helps reduce travel needs for monitoring and the associated costs. At the same time this approach ensures the random selection of samples for every parameter.

It is a good practice to employ oversampling at the design stage, not only to compensate for any attrition, outliers or non-response associated with the sample, but also to prevent a situation at the analysis stage where the required reliability is not achieved, and additional sampling efforts would be required. The sample size shown above will be adjusted upwards to account for non-responses, PP shall determine the appropriate non-responses rate based on previous experience.

(b) Data:

(i) Field Measurements:

To monitor the number of stoves that continue to be in use ($N_{y,i,j}$), the data collected will be a representative number of stoves in the database that are in use for the monitoring period. The method of collecting data will be field surveys of required sample size of ICS users in the database. Data will be collected from the field surveys, entered in the database and included in the monitoring report.

For monitoring $B_{y=1,new,i, survey}$, the data collected will be representative number of stoves in the database that were distributed within one year of start date of project implementation. The method of collecting data will be field survey and frequency will be once during the entire crediting period.

The table below summarizes field measurement data requirements

Parameter	Timing (indicative)	Frequency as per methodology	Methods to be applied	Comments on seasonal fluctuation
$N_{y,i,j}$	Monitoring will likely occur at least every 12 months	At least once in two years.	Visits to the end user premises, visual inspection and interview with ICS end-user	Unlikely to be due to any seasonal fluctuation.
$B_{y=1,new,i, survey}$	Within 1 st year from the starting		Visits to the premises, visual inspection,	Unlikely

	date of project implementation	Once. The value will be fixed for entire crediting period	measurement, and interview with ICS end-user.	
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(ii) Quality Assurance/Quality Control

The PP will apply measures to ensure the required confidence/precision for each sampled parameter is met, allowing for non-response and the possible removal of outliers from the sample, as part of a Quality Control/Quality Assurance system. The PP will determine the most effective measure for each parameter from the following list (illustrated using a required sample size of 20 and an effect of non-response of 2 to 4 ICS):

- Oversampling: Randomly draw a sample of minimum 24 ICS and collect data from each
- Buffer Group: Randomly draw a sample of at minimum 24 ICS and collect data from only 22 ICS. If this would not result in the required sample size data would be collected from the additional 2 ICS that were selected in the sample.
- Draw an additional sample: Randomly draw a sample of 22 ICS and collect data from these. If the required sample size is not achieved, an additional sample of 2 elements will be drawn and included in the sample.
- Use lower confidence bound (of $N_{y,i,j}$, $B_{y=1,new,i}$, survey) or, with a conservative approach according to the parameter definitions,

The PP may choose to stop monitoring a particular parameter once the required level of confidence/precision has been reached, as long as the calculated minimum number of samples has been achieved. As an example, the following steps could logically be followed for the case of applying a 30% buffer:

1. Visit first 10% of premises required for the 30% buffer. If the number of responses is sufficient to achieve the required reliability level, then stop sampling.
2. If step 1 is not sufficient to achieve the required reliability level, then visit the next 10% of premises (increases the additional sampling to 20% of the 30% buffer). If this additional sampling is sufficient, then stop sampling.
3. If step 2 is not sufficient to achieve the required reliability level, then complete the final 10% of the additional sampling buffer (bringing the total to 30%).

The sampling plan has the following procedures in place to ensure good quality data. The PP will ensure that field personnel have reviewed, understand and have signed the monitoring plan, including provisions for maximizing response rates, documenting out-of-population cases, refusals and other sources of non-response. A quality control and assurance strategy will be documented. Quality control and assurance strategies include addressing non-sampling errors, such as non-response or bias from interviewer. The monitoring plan will explain how to properly survey households to prevent bias from interviewer. In the case a household refuses to participate, another household will be chosen at random. To reduce interviewer bias, good questionnaire design and well-tested questionnaires will be used.

- (i) The calculation of the sample size will be carried out using estimates for parameter proportions, mean values and standard deviations, as the actual characteristics of the population/sampling frame are unknown. In order to ensure the quality of the sampling results, the PP can draw on the provisions for reliability calculations as provided by the

Guidelines for Sampling and Surveys in CDM Project Activities and Programme of Activities (version 04). In the event that the sampling results do not fulfil the required level of confidence and precision, the PP can undertake additional samples. If the reliability is still not sufficient after additional samples or other measures, the sampling may be repeated with an increased sample size. Alternatively, the PP may choose to apply the lower bound or higher bound according to the more conservative approach. Data archiving

Hard copies of the surveys will be kept and the registration database will have back up. Original stove purchase contracts or other means of acceptance by the users will be stored in the main office for the coordinating entity. A back-up of the registration database will also be stored on an electronic medium by the PP. All data monitored and required for verification and issuance will be kept for two years after the end of the crediting period or the last issuance of VCU for the project activity, whichever is later.

(ii) Analysis

The PP will manage a project database that includes the following data that can be directly attributable to the project activity, thereby allowing unambiguous determination of the emission reductions attributable the project:

- A list of households participating in project activity, including name, community/location, distribution date and unique serial number;
- Where replacements are made, assurance that the efficiency of the new ICS is similar to the specified.

Data obtained from the samples will be used to estimate proportions and mean values for the parameters described above. The values will then be factored into the emissions reduction calculations and result in the request for issuance of VCUs. The stoves that are not in use will be excluded from emissions reductions calculations and will not be counted towards the total number of ICS in operation during the monitoring period.

(b) Implementation

Sampling for the purpose of emission reduction calculation and elaboration of the monitoring report will occur at the end of each monitoring period. This sampling will be conducted by trained personnel either part of the project Implementer or PP team, or an experienced third-party entity. The maximum length of one monitoring period will be two years (duration, not calendar years). The project Implementer will be responsible for managing household data collection and entry into the project database. Field personnel will receive training on how to properly deal with surveying techniques and reduce errors. The project database will record the start and end dates of each monitoring period and record the emission reductions attributable to each monitoring period. Appropriate record keeping procedures will be implemented to ensure that each monitoring period data set can be transparently attributed to its corresponding project, preventing any occurrences of double counting. An internal review of the project database will be able to determine the status of project—the duration of previous monitoring periods, the households delivering monitoring data, and current verification activities.

Assessment for Leakage

According to methodology VMR0006, version 1.1, section 8.3; Leakage shall be considered as default 0.95 in accordance with Section 5.4 of AMS-II.G. therefore default value of leakage as 0.95 will be applied.

Monitoring Reporting

The PP will assess all monitoring data and produce one or two monitoring reports for the project for the VVB to verify corresponding to the preceding monitoring period the project. This report will present the data relating to the emission reductions generated by the project at the time of the monitoring period.

APPENDIX

References:

Appendix A: Local Stakeholder Consultation Details

Appendix B: List of Abbreviations

Appendix-A Local Stakeholder Consultation Details

Different mediums which were used for inviting stakeholder's opinion on the project

1. Newspaper Advertisement.

ADVERTISEMENT

The New Times 15

No. 5419, TUESDAY, DECEMBER 8, 2020

ANNOUNCEMENT FOR STAKEHOLDERS CONSULTATION MEETING - ENEDOM (RWANDA) - QUEST CAPITAL (CQC) KIGALI - RWANDA.

Introduction:

C-Quest Capital is a social impact investment enterprise based in Washington D.C., USA. We leverage carbon finance to provide the investment capital necessary for our stove programs to be carried out through partnerships with like-minded development organizations that use their networks of staff and trainers to reach the target households. To date, CQC through implementation partners has built over 750,000 stoves worldwide, with operational hubs in Malawi and Zambia, and have planned expansion efforts in 2021. Our investment program for 2021 is to provide an additional 3 Million Stoves over the next ten years across nine Sub-Saharan Countries.

At this end, we are working with ENEDOM (Energie Domestique) as our liaison and implementor for the first round of project implementation that will install 14,000 stoves by April 2021 before planning for next phases. (ENEDOM is a local company that is in the biomass technologies since 1997 in Rwanda). They have great experience in improved cookstoves, project design and implementation throughout Rwanda. ENEDOM are also assisting CQC with government coordination and implementation of this project as well as planning for stakeholders consultations in Rwanda which will take place 10th December 2020. All invited parties are encouraged to attend in person for the benefit of this project.

In case you have not received our invitation and interested to be part of this project, please contact our partner Mr. Kayunga Jean Marie Vianny tel: 0788501309 and e-mail: enedom@yahoo.fr.

Venue: EPD conference room- Kimihurura

Date: 10th December 2020

Time: 2:00-4:00 PM

Agenda

1. Registration
2. Introduction of participants
3. Welcoming Remarks
4. Objective of the project
5. Description of the project
6. Question and answers
7. Evaluation form
8. Closing of the meeting



ENEDOM
Managing Director
Kayunga Jean Marie Vianny

07/12/2020



National Bank of Rwanda
Banki Nkuru y'u Rwanda

Human Resources and Administration Directorate

Tender notices

The National Bank of Rwanda invites eligible bidders to submit their bids for the following tenders:

Sl ⁿ	Tender reference number	Title of tender
1	17/5/2020-21/OCB/C/BNR	Maintenance of generators
2	18/02/2020-21/OCB/C/BNR	Supply, installation and maintenance of UPS and provision of other related services

Tender documents may be obtained on any working day in the Procurement Division of the National Bank of Rwanda upon presentation of payment proof of a non-refundable fee of ten thousand Rwandan Francs (FRW 10,000) per tender to Account N°1266500010001 named "BNR Current Account" opened at National Bank of Rwanda.

Compulsory site visits for the tender for supply, installation and maintenance of UPS and provision of other related services are scheduled at BNR Headquarters on 15th December 2020 at 9:00 am, and at Mutanze and Rubavu Branches on 16th December 2020 at 10:00 am and 2:30 pm respectively. They will be conducted by BNR representatives.

All bids shall be accompanied by the required Bid Security of an amount indicated in the tender document in Rwandan Francs or equivalent in a convertible currency.

Well printed bids properly bound and presented in four copies one of which is the original must reach the Procurement Division of the National Bank of Rwanda, KN 6 Avenue, 4, Second Floor not later than 7th January 2021 at 09:30 am local time. Bids opening will be conducted on the same date in the presence of bidders or their representatives who choose to attend at 10:00 am local time in the meeting room. Late bids will be rejected and returned unopened. All envelopes should clearly indicate the tender title and the name of bidder.

Done at Kigali on 3rd December 2020.

BAZAMBARIZA Moise
Manager, Procurement Division

IRIMOZA Frances
Executive Director, Human Resources and Administration

National Bank of Rwanda-Restricted

1. Invitation Mail

Local Stakeholder Consultation Feedback Requested - Improved Cookstoves (Rwanda)



Brent Moser (bmoser@CquestCapital.com)
To: Brent Moser (bmoser@CquestCapital.com)
Cc: Jason Steele (jsteele@CquestCapital.com)

Reply Reply All Forward

Fri 12/18/2020 12:38 PM

Dear Madam / Sir:

As the Project Participant, C-Quest Capital would like to request local stakeholders in Rwanda to provide feedback on their proposed improved cookstove project for Rwanda. The project has been developed under VERRA/Voluntary Carbon Standard (VCS) and seeks to obtain Voluntary Carbon Units (VCUs). This program plans to disseminate and install high efficiency firewood cookstoves to poorer households in Rwanda that cook with open-fires with the goal to significantly reduce the dependency on Rwanda's remaining natural resources, reduce greenhouse gas emissions related to cooking, and contribute to improved health outcomes for the end-user. Carbon finance will be the primary source of funding for this program.

All program information including non-technical summary and project presentation is located on our website: <https://cquestcapital.com/blog/ssalisc>

You can provide feedback by replying to this email or through the feedback form on our website via the link above.

Please provide any feedback by **15 January 2021**.

Brent Moser
Director of Operations – Africa Region
C-Quest Capital LLC
1015 18th Street NW Suite 730
Washington, DC 20036
Phone +1 407.491.0368
Skype moser.brent
E-mail bmoser@cquestcapital.com
Website www.cquestcapital.com

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2. Invitation Letter from CQC-ENEDOM



ENEDOM

Energie Domestique

C-Quest Capital - ENEDOM

Tel: 0788501309

E-mail: enedom@yahoo.fr

To: Country Director,

GIZ- Rwanda

Dear Sir/Madam,

INVITATION CQC-ENEDOM 14,000 STOVE PROJECT CONSULTATION MEETING

Location: Energy Private Developers (EPD)-Kimiurura

Time: 2:00PM, Thursday 10th December 2020

C-Quest Capital-USA and ENEDOM (Energie Domestique)- Rwanda are developing clean cooking project aimed at installing 14,000 efficient stove in Rwanda by April 2020 as their first phase of bigger plan in the coming years. It is in this context that CQC-ENEDOM invites you for its first consultative meeting to explain our project to different stakeholders and gain inputs from bigger audience.

We believe that your contribution during this consultation will add value to the intended objective and benefits both sides to bring clear and reliable source of cooking for the people.

The agenda is outlined below. And we remind the invitees of the current COVID 19 Pandemic outside there, everyone is required to respect the ministry of health guidelines and we will facilitate in keeping it at all time at the venue.

Yours sincerely,

C-Quest Capital - ENEDOM



Agenda

- 2:00 PM. Arrival and Registration
- 2:15 PM. Introduction of participants
- 2:20 PM. Welcoming Remarks
- 2:25 PM. Objective of the project
- 2:30 PM. Description of the project
- 3:00 PM. Question and answers

3. List of Attendees.

Local Stakeholder Consultation for Local Stakeholder Consultation for Installation of High Efficiency Firewood Cookstoves using VERRA/Voluntary Carbon Standard (VCS)

PARTICIPANTS LIST

No.	Name	Organization / Community	Position in Org. / Community	Physical Address	Email Address	Phone Number	Signature
1	Timothy Kagame	Practical Action	Energy Tech Specialist	Kigali	timothy.kagame@practicalaction.or.ke	0738466635	[Signature]
2	KAYOUSA Jean Marie	ENERDON	MD	Kigali	enerdon@kigali.rw	0788301309	[Signature]
3	MAHISTANA ANDU	SNV	Energy Sector Manager	Kigali	maahistana@snv.org	0788301309	[Signature]
4	Josephine Mubanzi	EPD	Comm & Policy	Kigali	josephine.mubanzi@epd.rw	0788301309	[Signature]
5	Rusomon Patrick	RIB	State Officer	Kigali	patrick.rusomon@rib.rw	0788301309	[Signature]
6	Ruhumunzi Eric	EPD	Data Analyst	Kigali	eric.ruhumunzi@gmail.com	0788301309	[Signature]
7	NIYIBIZI Aham	REG/EDCL	ICS Engineer	Kigali	aham.niyibizi@edcl.rw	0788301309	[Signature]
8	UNYONYE Jean Claude	EPD / Chair Clean Cooking and Biomass Sub-Sector	MD RECONC Ltd	Kigali	unyonye.jc@yahoo.com	0788301309	[Signature]

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Local Stakeholder Consultation for Local Stakeholder Consultation for Installation of High Efficiency Firewood Cookstoves using VERRA/Voluntary Carbon Standard (VCS)

PARTICIPANTS LIST

No.	Name	Organization / Community	Position in Org. / Community	Physical Address	Email Address	Phone Number	Signature
9	MININFRA	Steven	Primary Specialist	Kigali	steven.bihindye@mininfra.gov.rw	0788301309	[Signature]
10	Kamanzi Emmanuel	Consultant	Consultant	Kigali	kamanze2@yahoo.com	0788301309	[Signature]

List of Electronic Feedback Providers

No.	Organization	Name	Title	Email Address	Date Received
1	Practical Action	KAMPIRE Berthille	Field Coordinator Gihembe Refugee Camp	Bertill.kampire@practicalaction.or.ke	18-Dec-20
2	GIZ RW & EnDev	Isabella Troconis	Component Coordinator	Isabella.troconis@giz.de	11-Dec-20
3	REMA	Oreste Niyonsaba Herman Hakuzimana	Directors of Primary & Alt. Energies at REMA	oniyonsaba@edcl.rw; hakuzimana@rema.org	14-Dec-20

5 Photographs.



Appendix B- List of Abbreviations:

CDM: Clean Development Mechanism

CQC: C-Quest Capital LLC

GHG: Green House Gas

ICS: Improved Cookstoves

PP: Project Proponent

SDG: Sustainable Development Goals

SD VISta: Sustainable Development Verified Impact Standard

TLC: Total Land Care

VCS: Voluntary Carbon Standard

VCU: Voluntary Carbon Unit